Kevin Zhang

kevinkw.zhang@mail.utoronto.ca • diebrate.github.io • +1-647-896-6853
LinkedIn: kevin-zhang-uoft • GitHub: Diebrate • Citizenship: Canada

Research interests

Optimal transport, stochastic differential equations, machine learning, artificial intelligence, scRNA-seq, generative modelling, anomaly detecion, game theory

Education

2019 – Present **University of Toronto** – Toronto, Ontario, Canada
PhD in Statistics
Advisors: Dr. Dehan Kong, Dr. Zhaolei Zhang. *GPA: 3.8.*

2015 – 2019 **University of Toronto** – Toronto, Ontario, Canada
HBSc in pathobiology, statistics and mathematics. *GPA: 3.89.*

Research experience

2023 – Present **Generative Modelling Using Zero-Shot Learning and Transfer Learning (ongoing project)**
Supervisors: Dr. Dehan Kong (University of Toronto), Dr. Zhaolei Zhang (University of Toronto).
Currently designing novel numerical techniques that perform transfer learning among multiple batches of data (possibly embedded in different manifolds)
Making connections between transfer learning and classical statistical models (e.g., regression, classification).

2020 – Present **Trajectory Inference Using Stochastic Differential Equations and Mean Field Control (manuscript submitted and in revision (PLOS Computational Biology), preprint available)**

Supervisors: Dr. Dehan Kong (University of Toronto), Dr. Zhaolei Zhang (University of Toronto).

Proposed a comprehensive mathematical diffusion model integrating concepts from mean field control and SDE to capture crowd motion in a population observed over a timeline.

Constructed a neural network using PyTorch to parametrize the SDE model.

Implemented an innovative numerical algorithm more suitable for models with higher complexity as compared to other existing SDE solvers.

Analyzed multiple different data sets and demonstrated excellent customizability and performance from the model.

2021 – 2023 **Modelling Single-Cell Developmental Trajectory Using Optimal Transport (manuscript in preparation)**

Supervisors: Dr. Dehan Kong (University of Toronto), Dr. Zhaolei Zhang (University of Toronto).

Proposed a novel test statistic for change point detection in time series data and constructed an innovative mathematical model for population movement.

Developed an integrative framework written in Python, R and MATLAB for numerical optimization and performance evaluation.

The proposed testing method outperformed other state-of-the-art methods in change point detection.

Analyzed high-dimensional ($\approx 20,000$ dimensions) time series data from over 250,000 cells.

September 2017 – April 2018 **Mount Sinai Hospital - Kandel Lab (Project Student)**

Supervisor: Dr. Rita Kandel (University of Toronto.

Analyzed effects of Tgfb3 on the regeneration of nucleus pulposus tissue (cell/tissue culture and extracellular matrix quantification).

May 2017 – August 2017 **MaRS Discovery District - Mekhail Lab (Summer Student)**

Supervisor: Dr. Karim Mekhail (University of Toronto).

Analyzed **effects of Poly-Q expansion on yeast replicative lifespan**(RNA variants and cell culture).

Publications

2023 **Modeling Single Cell Trajectory Using Forward-Backward Stochastic Differential Equations**
K. Zhang, J. Zhu, D. Kong, and Z. Zhang.
PLOS Computational Biology, (under revision).[Link to preprint]Link to GitHub

2023 **Modeling Cell Type Development Trajectory using Multinomial Unbalanced Optimal Transport**
K. Zhang, J. Zhu, D. Kong, and Z. Zhang.
Manuscript in preparation.Link to GitHub

2023 **LLOT: application of Laplacian Linear Optimal Transport in spatial transcriptome reconstruction**
J. Zhu, K. Zhang, D. Kong, and Z. Zhang.
Manuscript submitted.

2018 **Conserved pbp1/ataxin-2 regulates retrotransposon activity and connects polyglutamine expansion-driven protein aggregation to lifespan-controlling rdna repeats.**
L. Ostrowski, A. Hall, K. Szafranski, R. Oshidari, K. Abraham, J. Chan, C. Krustev, K. Zhang, A. Wang, Y. Liu, R. Guo, and K. Mekhail.
Communications Biology.

Teaching experience

Fall 2019-Present **University of Toronto (Teaching Assistant** – Toronto, Ontario, Canada
Courses include:
STA130: An Introduction to Statistical Reasoning and Data Science
STA220: The Practice of Statistics I
STA255: Statistical Theory
STA261: Probability and Statistics II
STA302: Methods of Data Analysis I
STA303: Methods of Data Analysis II
STA304: Surveys, Sampling and Observational Data
STA305: Design and Analysis of Experiments
STA347: Probability
STA437: Methods for Multivariate Data
STA442: Methods of Applied Statistics
Held weekly tutorials/office hours and monitored online course forums. Invigilated and marked midterms and exams.

Work experience

2019 - Present **University of Toronto (Graduate Research Assistant)** – Toronto, Ontario, Canada
Worked under the supervision of Dr. Dehan Kong and Dr. Zhaolei Zhang.
Research topics include applications of optimal transport and stochastic differential equations in machine learning with a focus on anomaly detection and generative modelling in time series data (specifically single-cell RNA sequencing data).

2019 - Present **University of Toronto (Teaching Assistant)** – Toronto, Ontario, Canada
Responsible for various undergraduate statistics courses from introductory probability theory to advanced multivariate statistics.
Held tutorials as mini-lectures for undergraduate students.

Review experience

2023 Canada Communicable Disease Report (2 times)

2023 Journal of the American Statistical Association (1 time)

2022 Journal of the Royal Statistical Society: Series C (1 time)

Talks and conferences

December 2022 Modelling Cellular Development Trajectory using Unbalanced Optimal Transport
Medicine by Design's 7th Annual Symposium

July 2022 Modelling Single-Cell Type Trajectory Using Optimal Transport
The Fifth ICSA-Canada Chapter Symposium

Honors and scholarships

2015-2019 Dean's List (University of Toronto)

2018 New College Council In-Course Scholarship (University of Toronto)

2017 LMP Undergraduate Summer Research Program UROP Award (University of Toronto)

2015 Undergraduate Entrance Award (University of Toronto)

Technical skills

Programming languages
Proficient in: Python, R, SQL
Familiar with: C++, Linux/UNIX environment, MATLAB, HTML

Libraries/Packages
Python: NumPy, Pandas, matplotlib, PyTorch (including CUDA GPU Configuration), TensorFlow, scikit-learn, XGBoost
R: dplyr, ggplot2, tidyverse

Software
LaTeX, Git, Microsoft Office, SLURM Workload Manager

Languages
English (fluent), Chinese (native)